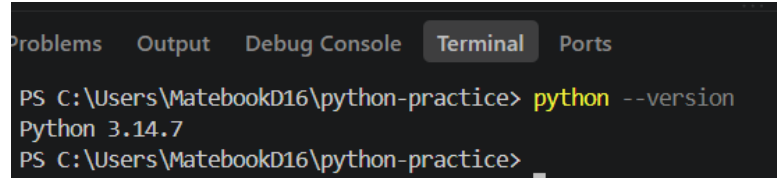


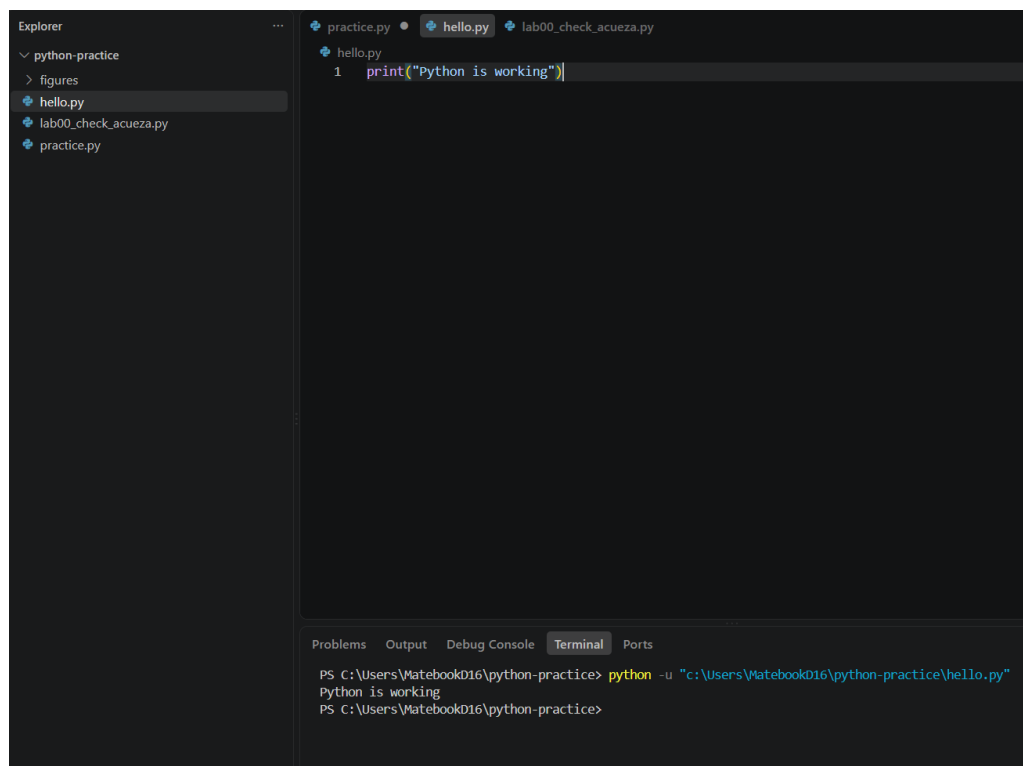
Assignment #0

1. The terminal showing python --version and its answer.



```
Problems Output Debug Console Terminal Ports
PS C:\Users\MatebookD16\python-practice> python --version
Python 3.14.7
PS C:\Users\MatebookD16\python-practice>
```

2. VS Code with hello.py open and its output visible in the terminal.

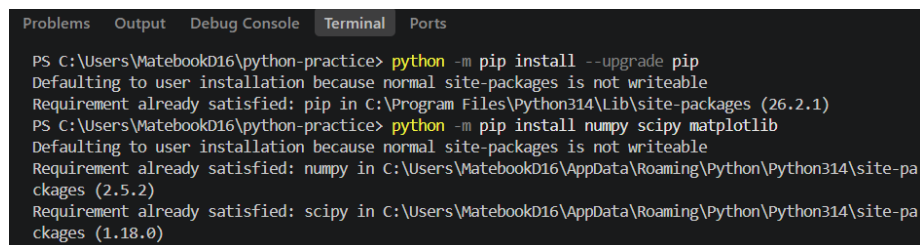


```
Explorer
python-practice
  figures
    hello.py
    lab00_check_acueza.py
    practice.py

practice.py hello.py lab00_check_acueza.py
hello.py
1 print("Python is working")

Problems Output Debug Console Terminal Ports
PS C:\Users\MatebookD16\python-practice> python -u "C:\Users\MatebookD16\python-practice\hello.py"
Python is working
PS C:\Users\MatebookD16\python-practice>
```

3. The terminal showing the successful installation of NumPy, SciPy and Matplotlib.



```
Problems Output Debug Console Terminal Ports
PS C:\Users\MatebookD16\python-practice> python -m pip install --upgrade pip
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: pip in C:\Program Files\Python314\Lib\site-packages (26.2.1)
PS C:\Users\MatebookD16\python-practice> python -m pip install numpy scipy matplotlib
Defaulting to user installation because normal site-packages is not writeable
Requirement already satisfied: numpy in C:\Users\MatebookD16\AppData\Roaming\Python\Python314\site-packages (2.5.2)
Requirement already satisfied: scipy in C:\Users\MatebookD16\AppData\Roaming\Python\Python314\site-packages (1.18.0)
```

```
Problems Output Debug Console Terminal Ports

Requirement already satisfied: matplotlib in C:\Users\MatebookD16\AppData\Roaming\Python\Python314\site-packages (3.11.1)
Requirement already satisfied: contourpy>=1.0.1 in C:\Users\MatebookD16\AppData\Roaming\Python\Python314\site-packages (from matplotlib) (1.3.3)
Requirement already satisfied: cycler>=0.10 in C:\Users\MatebookD16\AppData\Roaming\Python\Python314\site-packages (from matplotlib) (0.12.1)
Requirement already satisfied: fonttools>=4.28.2 in C:\Users\MatebookD16\AppData\Roaming\Python\Python314\site-packages (from matplotlib) (4.63.0)
Requirement already satisfied: kiwisolver>=1.3.1 in C:\Users\MatebookD16\AppData\Roaming\Python\Python314\site-packages (from matplotlib) (1.5.0)
```

```
Problems Output Debug Console Terminal Ports

n314/site-packages (from matplotlib) (1.5.0)
Requirement already satisfied: packaging>=20.0 in C:\Users\MatebookD16\AppData\Roaming\Python\Python314\site-packages (from matplotlib) (26.3)
Requirement already satisfied: pillow>=9 in C:\Users\MatebookD16\AppData\Roaming\Python\Python314\site-packages (from matplotlib) (12.3.0)
Requirement already satisfied: pyparsing>=3 in C:\Users\MatebookD16\AppData\Roaming\Python\Python314\site-packages (from matplotlib) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in C:\Users\MatebookD16\AppData\Roaming\Python\Python314\site-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in C:\Users\MatebookD16\AppData\Roaming\Python\Python314\site-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)
PS C:\Users\MatebookD16\python-practice>
```

```
Problems Output Debug Console Terminal Ports

Requirement already satisfied: pillow>=9 in C:\Users\MatebookD16\AppData\Roaming\Python\Python314\site-packages (from matplotlib) (12.3.0)
Requirement already satisfied: pyparsing>=3 in C:\Users\MatebookD16\AppData\Roaming\Python\Python314\site-packages (from matplotlib) (3.3.2)
Requirement already satisfied: python-dateutil>=2.7 in C:\Users\MatebookD16\AppData\Roaming\Python\Python314\site-packages (from matplotlib) (2.9.0.post0)
Requirement already satisfied: six>=1.5 in C:\Users\MatebookD16\AppData\Roaming\Python\Python314\site-packages (from python-dateutil>=2.7->matplotlib) (1.17.0)
PS C:\Users\MatebookD16\python-practice>
```

4. The full console output of your verification program, ending in PLATFORM READY.

```
practice.py hello.py lab00_check_acueza.py
lab00_check_acueza.py > ...
1 """
2 Lab Assignment 0 - platform verification
3 Name : Nicah Jean B. Acueza
4 Student number : TUPM-24-0451
5 """
6 import os
7 import sys
8
9 import numpy as np
10 import scipy
11 from scipy import integrate
12 import matplotlib
13 import matplotlib.pyplot as plt
14
15 # ---- 1. report the versions ----
16 print("Python :", sys.version.split()[0])
17 print("NumPy :", np.__version__)
18 print("SciPy :", scipy.__version__)
19 print("Matplotlib :", matplotlib.__version__)
20
21 # ---- 2. a NumPy calculation ----
22 # 200 points from 0 to 2*pi, then the sine of every one of them at once.
23 x = np.linspace(0, 2*np.pi, 200)
24 y = np.sin(x)
25 print("\nmean of sin(x) over one period :", round(float(y.mean()), 12))
26
27 # ---- 3. a SciPy cross-check ----
28 # The integral of sin(x) from 0 to pi is exactly 2.
29 area, _ = integrate.quad(np.sin, 0, np.pi)
30 print("integral of sin(x) from 0 to pi :", round(area, 12), "(exact value 2)")
31
32 # ---- 4. a Matplotlib figure ----
33 os.makedirs("figures", exist_ok=True)
34 plt.rcParams.update({"figure.dpi": 150})
35
36 fig, ax = plt.subplots(figsize=(5.2, 3.2))
37 ax.plot(x, y, color="#0b5fa5", lw=1.6, label="sin(x)")
38 ax.plot(x, np.cos(x), color="#8a5b00", lw=1.2, ls="--", label="cos(x)")
39 ax.set_xlabel("x (radians)")
40 ax.set_ylabel("value")

lab00_check_acueza.py > ...
C:\Users\MatebookD16> python-practice > lab00_check_acueza.py
9 import numpy as np
10 import scipy
11 from scipy import integrate
12 import matplotlib
13 import matplotlib.pyplot as plt
14
15 # ---- 1. report the versions ----
16 print("Python :", sys.version.split()[0])
17 print("NumPy :", np.__version__)
18 print("SciPy :", scipy.__version__)
19 print("Matplotlib :", matplotlib.__version__)
20
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35
36 fig, ax = plt.subplots(figsize=(5.2, 3.2))
37 ax.plot(x, y, color="#0b5fa5", lw=1.6, label="sin(x)")
38 ax.plot(x, np.cos(x), color="#8a5b00", lw=1.2, ls="--", label="cos(x)")

Problems Output Debug Console Terminal Ports

SciPy : 1.18.0
Matplotlib : 3.11.1

mean of sin(x) over one period : 0.0
Integral of sin(x) from 0 to pi : 2.0 (exact value 2)

figure written to figures/lab00_check.png
PLATFORM READY
PS C:\Users\MatebookD16>
```

5. Your GitHub repository page, showing the repository name and the uploaded file.

